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Raman scattering studies of graphene under high pressure

Denis Machon,[a] Colin Bousige,[a] Rafael Alencar,[a,b] Abraao Torres-Dias,[a,b] Félix Balima,[a,c] Jimmy Nicolle,[a,d] Gardênia de Sousa Pinheiro,[a,e] Antonio Gomes Souza Filho[b] and Alfonso San-Miguel[a]*

The effect of pressure – by definition a 3-dimensional concept – on 2-dimensional systems brings a number of fundamental questions, which have been partly answered through newly engineered samples and experimental setups. In this review of the high-pressure Raman studies of graphene, we will in particular underline the importance of the presence of a supporting substrate and its role for the production of biaxial strain conditions in high pressure experiments. Raman shifts observed during these experiments may be related to the strain induced by the substrate rather than by the applied pressure, graphene stress, and hydrostatic pressure not being straightforwardly equivalent in this case of a composite 2D-on-3D material. We will also consider the effects of the nature of the substrate surface, the nature of the pressure transmitting media, as well as the effect of the number of layers on the high-pressure response of the 2D–system, by using the phonon behavior as strain probe as pressure is applied. The effects of adhesion and of strain transfer from the substrate to the 2D–system will be at the heart of our discussion. **Copyright © 2017 John Wiley & Sons, Ltd.**

Keywords: graphene; high pressure; strain; 2D–systems

The mechanical response of graphene and Raman spectroscopy

Graphene may be considered as one of the simplest nano-objects but in spite of that, it raises many different scientific and technological challenges. The mechanical response of graphene is one of them, the main reason being that the continuum mechanics description of the elasticity of 3D bodies cannot be applied to graphene due to its single atom thickness structure. Because pressure is a force exerted on a surface, how is the environment hydrostatic pressure transmitted to a 2D material? The lack of a definition for what could be the graphene thickness[1] indeed constitutes a barrier for the characterization of applied stresses, and the 3D description of continuum mechanics needs to be adapted to the 2D case. One way of tackling the problem is to consider graphene as the bidimensional (2D) elemental building block of graphite[2]. We could then think of the graphene mechanical properties as derived from the graphite ones by just reducing the out of plane dimension. Ultimately, the c-axis of graphite may not be considered anymore. We would then end up with an elasticity tensor description having just two elastic constants (C_{11} and C_{12}) and an isotropic elastic response in the plane, like for all hexagonal systems[3,4] – the usual additional elastic constants C_{33} and C_{13} when considering a 3D sample not having any significance anymore. A central question arises from these considerations: Would in that case the corresponding C_{11} and C_{12} graphite elastic constants be conserved for graphene when making such dimension reduction? Graphene evolves anyway in a 3D world, and one of the most convenient ways to tackle graphene mechanics is thus to consider the thin-plate model, also referred as the Kirchhoff–Love Theory.[5] We end up then with a three independent elastic constants problem that can be used to formulate variables as the bending stiffness, D, or the in-plane stiffness, C, which is the 2D analog of the Young modulus for 3D objects. Whatever the model used, high pressure application on graphene could lead to modifications of the in-plane carbon bonding through the interaction with the environment atoms or molecules. This could then consequently have an effect on the corresponding elastic constants. In fact, pressure-induced bonding modification have been recently proposed for water as pressure transmitting medium.[6]

Raman scattering, through its sensitivity to highly symmetric covalent bonds with little or no natural dipole, has proven to be an excellent probe for carbon materials since practically the beginning of the use of laser sources in Raman spectroscopy.[7,8] In special, the carbon bonds with sp^2 hybridization being in-plane, Raman spectroscopy thus constitutes a natural probe

* Correspondence to: Alfonso San-Miguel, Univ. Lyon, Université Claude Bernard Lyon 1, CNRS, Institut Lumière Matière, F-69622, Villeurbanne, France. E-mail: alfonso.san-miguel@univ-lyon1.fr

a Univ. Lyon, Université Claude Bernard Lyon 1, CNRS, Institut Lumière Matière, F-69622 Villeurbanne, France

b Departamento de Física, Universidade Federal do Ceará, P. O. Box 6030, Fortaleza, Ceará, 60455-900 Brazil

c ICMCB – CNRS, UPR9048, 3 Avenue Pey Berland, 33600 Pessac cedex, France

d Laboratoire Interface Confinement Matériaux et Nanostructures, UMR 7374, 1b rue de la Ferrolerie, 45071 Orléans Cedex 2, France

e Departamento de Física, Campus Ministro Petrônio Portella, Universidade Federal do Piauí, Teresina, Piauí, 64049-550 Brazil

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for the in-plane mechanics of graphene, and in particular through the monitoring of the G-band optical phonons, the latter being an excellent probe of the evolution of the interatomic potential under strain. Finally, in the case of high pressure studies, Raman spectroscopy is perfectly adapted thanks to the transparency of the diamond crystals used in diamond anvil cells (DAC) pressure apparatus. In this setup, the diamonds act as optical windows for most laser energies – the only limitation is the diamond UV absorption. One of the limitations of the DAC setup is nevertheless the partial overlap of the Raman prominent features of diamond (first order E_{2g} peak and its second order at 1300 and 2600 cm^{-1}, respectively) with those of graphene [defect-activated D-band and the 2D (G′) band]. This problem can however partially be reduced by the use of Raman confocal systems. In the other hand, the wavenumber of G-band in sp^2 network is 250 cm^{-1} higher than the Raman peak from diamond, thus, the measurement of this mode in DAC setup is straightforward.

There have been many different studies on the mechanical properties of graphene. Those studies can be classified in (1) out-of-plane (OP) strain experiments and (2) in-plane (IP) stress experiments. OP experiments require the suspension of the graphene plane above a trench or a hole, whereas this is not needed for IP experiments. Different methods (Fig. 1) have been used in OP experiments, such as the application of potential differences between the graphene plane and a gate,[9] the use of a gas pressure difference between both sides of graphene,[10] or the use of atomic force microscopy (AFM) or indentation devices.[11,12] IP experiments can be classified in uniaxial and biaxial stress configurations. In most cases, such experiments are based on the interaction of the graphene plane with a substrate that is deformed – thermally,[13] mechanically,[14,15] or electrically[16] in the case of piezo-electric substrates. In all these types of experiments, Raman spectroscopy can be used as a probe of the graphene mechanical response, but the in-plane response of the G-band makes it better adapted for IP experiments. Another important parameter is the type of mechanical response. In fact, uniaxial strain experiments will lead to the splitting of the graphene degenerate E_{2g} optical mode (the G-band splits into G^+ and G^-),[14] while this is not the case in biaxial strain.[17] As we will discuss later, in high pressure experiments, we will be concerned with biaxial strain. We may cite that other experimental methods can be used to explore the mechanical properties of graphene, as it has been recently shown in the determination of the bending modulus of graphene through the study of the mechanical response of carbon nanotubes of different diameters.[18]

Whatever the experimental probe used, the adhesion of graphene to its substrate constitutes an important parameter to be determined owing the impact it might have in the physical properties and device performance .[19,20] It will define the limits of both OP and IP experiments that can be determined by the advent of sliding or the transfer of stress from the substrate to graphene, respectively.

The accessible strains are one order of magnitude higher in 2D than in 3D materials. In fact, for 2D materials, the mechanical properties are close to the ones predicted by Griffith for fragile materials for which the failure point is dominated by the interatomic forces but not by defects as it is in the case in 3D materials.[21] This outstanding stretchability, i.e. the ability to sustain high strain without failure, opens up new opportunities unattainable for bulk materials and allows anticipating huge effects on physical properties.

How is the strain transferred between the substrate and the deposited 2D material is a challenging question that is important to answer before using stress as a tuning parameter of the 2Ds physical properties. One way to maximize substrate-graphene adhesion is by pressure application. In fact, pressure will reduce very effectively the carbon-substrate distance increasing then the interaction between the two subsystems. This has been for instance shown on gold contacted carbon nanotubes submitted to pressure in which the evolution of that distance was found to be comparable with the evolution of the graphene–graphene distance in graphite under pressure.[22] This pressure induced distance evolution is much more important than the in-plane carbon–carbon distance.[23]

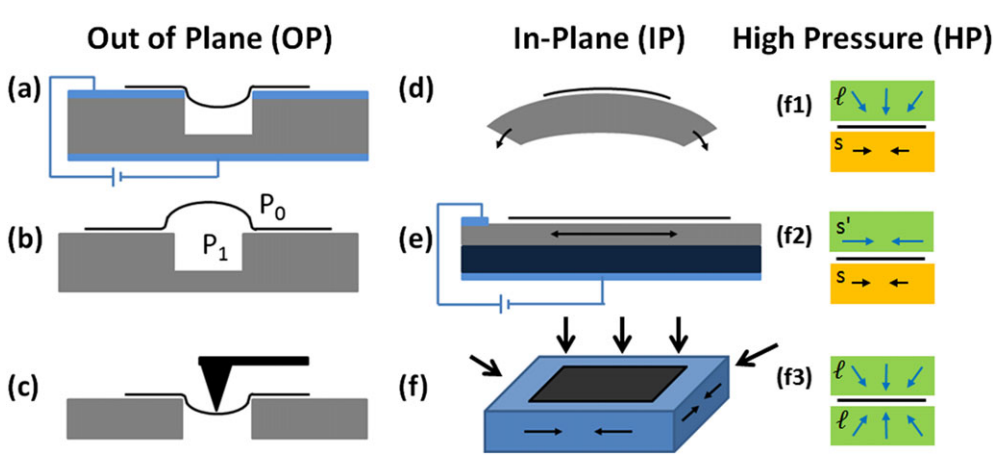

Figure 1. Schemes of possible mechanical probe experiments performed on graphene. Most of these methods have been combined with Raman spectroscopy. We distinguish two families of experiments: (1) Out-of-plane experiments (OP) in which a polarization tension (a), a difference of pressures (b), or an atomic force microscopy mechanical experiment can be used to deform the graphene membrane out of its plane. (2) In-plane experiments (IP) in which graphene can be deformed uniaxially or biaxially. The use of a mechanically deformed substrate (d), a piezo-actuator (e), or the biaxial compression of the substrate using and external high pressure (f) are examples of such methods. The different schemes of high pressure experiments are detailed in f1–f3. Graphene can be supported (f1, f2) by a solid (s) and embedded in a liquid (f1) or a solid s′ (f2) pressure transmitting medium. Nonsupported graphene in a liquid pressure transmitting medium (ℓ) is also shown (f3), a configuration that can appear in the case of unbinding of the 2D sample from its substrate.

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In Fig. 2 is depicted a typical setup for the study of supported graphene under high pressure. In general, a liquid pressure transmitting medium can be used but at a given pressure, it will solidify.[24] In a same experiment, the graphene sample can then be found in the configurations f1 and f2 as shown in Fig. 1 and even in configuration f3 if unbinding between substrate and graphene and subsequent invasion of the pressure transmitting medium in the interfacial region takes place. In an adhesive configuration (Fig. 1 f1 and f2), the most significant effect on the system will be the volume reduction of the substrate governed by its equation of state. Such volume reduction will lead to a biaxial compression of graphene (or any 2D system in adhesion).[25] Coupled with the enhanced pressure induced adhesion, this experimental setup allows for biaxial compressive strain excursions difficult to attain by any other means.

The investigation of the graphene mechanical properties is of extreme importance in the design of new graphene-based nanocomposites[26] or the development of electronic or mechatronic devices.[27] As will be discussed later, high pressure investigation of graphene constitutes also a formidable benchmark test for the investigation of graphene adhesion under extreme mechanical efforts and can also provide important information in the domain of tribology as well as in operating devices at extreme conditions.

In this review, we will discuss the contribution of high-pressure experiments to the understanding of both graphene mechanical properties and its interaction with its environment. The manuscript will be organized as follows: in section II, we will make a survey of the high pressure experiments on graphene. We will then consider the effect of the nature of the substrate on the high pressure response of supported graphene in section III, the effect of the number of layers in section IV, and the effect of the pressure transmitting medium in section V. Finally, section VI will provide final discussion and concluding remarks.

Graphene under extreme conditions – a survey

After the isolation of graphene in 2004, many systematic studies of fundamental issues were carried out by applying electric fields, magnetic fields, strain, pressure, doping, substrate effects[17,28] to investigate the physical properties of this 2D model material. From a mechanical properties point of view, the effects of strain through high pressure studies were first reported in 2009 by Proctor *et al.*,[29] followed by Nicolle *et al.*[30] These works were carried out on graphene samples sitting on Si/SiO_2 substrates. As said before, the main advantages of high pressure studies are that large compressive strain values can be achieved and that the coupling between graphene and the supporting substrate – and thus the interfacial strain transfer – are maximized.

A first finding of these studies is that the pressure-dependence of the G-peak in graphene is different from the one observed in graphite. The G-band upshift with pressure is characterized by its slope $\partial\omega_G/\partial P$, and this parameter was found to be significantly high for graphene (~16 $cm^{-1}.GPa^{-1}$[29] and ~8 $cm^{-1}.GPa^{-1}$[30]) as compared with graphite (~4.7 $cm^{-1}.GPa^{-1}$).[23] A subsequent study investigated the role of the substrate and of the pressure-transmitting medium (PTM)[31] using graphene grown by chemical vapor deposition on a Cu substrate. Here again, the slope $\partial\omega_G/\partial P$ was found around ~9 $cm^{-1}.GPa^{-1}$ whatever the PTM used, thus discarding the effects of charge transfer on the pressure-induced evolution of the G-peak, contrary to what had been previously proposed.[30] A recent work using various substrates (diamond and sapphire) showed a strong dispersion of the G-band slope values with the substrate.[32] Surprisingly, the value for graphene on diamond is even lower than the graphite G-band slope (Table 1).

A second interesting point reported in reference[30] is that a larger slope of $\omega_G = f(P)$ is found for a single layer and a bilayer, but not for a trilayer graphene or multilayer graphene. For the latter,

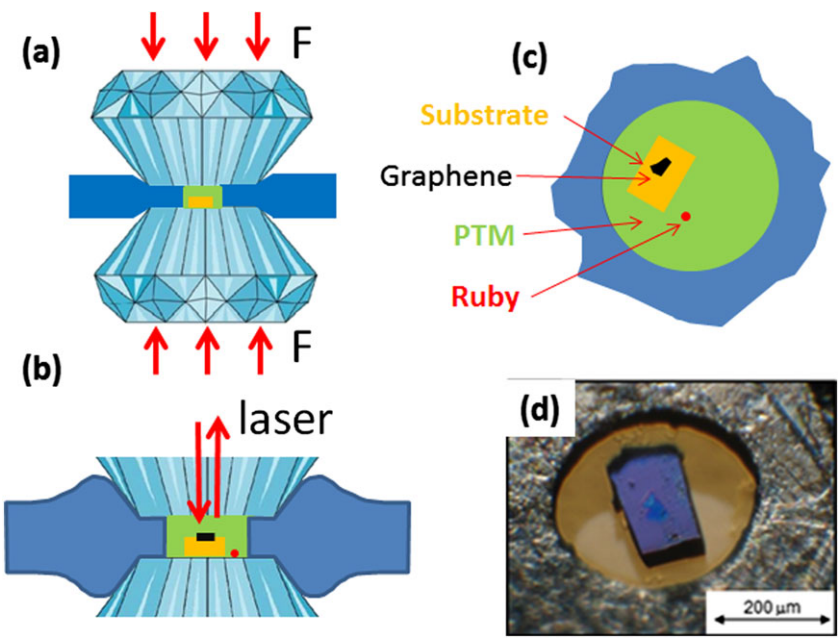

Figure 2. Principle of Raman experiments under high pressure on supported graphene. (a) Scheme of a diamond anvil cell. Two opposite diamonds define together with a drilled gasket, an experimental cavity that will receive the substrate/graphene sample which together with a ruby chip used to calibrate the pressure will be embedded in a pressure transmitting medium (PTM). (b and c) Detail of the loading scheme in two perpendicular projections. (d) Picture of a typical loading where the graphene sample is sitting on a Si/SiO_2 substrate.

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Table 1. Pressure dependency of the G-peak of a graphene sample on different substrates having linear compressibility β_S

Substrate	$\partial\omega G/\partial P$ ($cm^{-1}.GPa^{-1}$)	β_S (GPa)	PTM	Ref
Diamond	4.0 ± 0.2	1329	Alcohol	[32]
Sapphire	7.8 ± 0.5	750	Alcohol	[32]
Copper	9.2 ± 0.3	420	Alcohol/Fluorinet	[31]
SiO_2	10.5 ± 0.2	114	Alcohol, N_2	[30]This work

For all substrates, graphene samples were CVD-grown (and transferred for diamond and sapphire) except for the SiO_2 in which is exfoliated. For the pressure transmitting media (PTM), the 4 : 1 methanol : ethanol mixture is referred as 'Alcohol' and the FC70 : FC77 Fluorinert mixture as 'Fluorinert'; CVD, chemical vapor deposition.

the pressure-dependency, as expected, is identical to the case of graphite.

Tentative measures of freestanding, i.e. nonsupported, graphene under high pressure have been reported. This includes the work of Proctor *et al.*[29] in which liquid-phase exfoliation of graphite was performed. Their dispersion included mixtures of monolayer, bilayer, and few-layer graphene and also some nanographite pieces. However, the particularity of the Raman spectra that includes a strong D-band makes any interpretation hazardous.[27] In the work of Filintoglou *et al.*,[31] in some cases, a sudden and irreversible change of behavior of the G-band evolution of graphene supported on copper is observed. This result is interpreted as a consequence of the detachment of graphene from the substrate it is sitting on, thus leading to a G-band slope of 5.6 cm^{-1}/GPa, typical of an unsupported graphene, as compared with 9.1–9.3 cm^{-1}/GPa when graphene is in the adhesion regime. We may note that this value is in rather good agreement with the one of Hadjikhani *et al.*[33] of an approximate slope of 5 cm^{-1}/GPa also for chemical vapor deposition graphene grown on copper foil. In this last work, 4 : 1 methanol : ethanol was used as pressure transmitting medium. When using 4 : 1 methanol : ethanol, however, Filintoglou *et al.*[29] did not observe the detachment of the graphene sheet. These differences can then be interpreted by different states of graphene-substrate adhesion, which could include other parameters such as the surface roughness or the presence of defects.

All these experimental results will be detailed and tentatively interpreted in a coherent framework in the following sections of the manuscript.

Beside the mechanical properties of graphene, other aspects have been explored in high pressure experiments from which we can mention the evolution of the van Hove singularities,[34] or the coupling evolution between graphene layers.[35]

High-pressure response of graphene: effect of the substrate

As discussed in the introduction, the strain transmission in 3D materials is well understood by using the continuum mechanics laws, but this description fails for 2D materials. Thus, let us consider the case of pressure, which is a force divided by a surface: How pressure can be defined and transmitted laterally on a sample that shows an atomic thickness and for which the concept of lateral surface is irrelevant?

Usually, in a high-pressure experiment performed on a sample studied by Raman spectroscopy, the Raman shift is plotted as a function of pressure. The implicit assumption is that the stress is totally transferred to the sample and that pressure is then a relevant parameter. This is correct until the appearance of nonhydrostatic stress components. Then, the nature of the stress is often discussed to explain some peculiarities in the ω(P) plots. In the case of 2D materials in general and for graphene in particular, Raman experiments at high pressure (i.e. under biaxial stresses) are summarized by plotting the same graphs, i.e. Raman shift as a function of pressure – as if the hydrostatic stress were totally transmitted until the observation of anomalies that are interpreted, for instance, as the slipping of the graphene sheet over the substrate. The physical properties of graphene (phonons, bandgap, etc.) versus pressure were determined on this basis. However, this idea remains to be clearly demonstrated, and it is under debate. For instance, in case of experiments using uniaxial stress by deformation of a substrate under tension, the basic assumption is that it is the strain that is transmitted. In addition, the effect of the substrate must be further explained. Understanding the pressure-induced evolution of graphene requires considering the system (substrate + 2D material) and the adhesion between the two subsystems. Differences in adhesion will result in different pressure dependencies of the Raman peaks (Table 1).[32]

#Such differences are surprising considering that the same chemical C-C bond is probed in each case for the same applied pressure. Therefore, let us consider the physical effects at the length scale of the chemical bond, having in mind that the Raman shift is attributed to the bond length variation. In a high-pressure experiment, the main starting hypothesis is that the *strain* is transmitted from the substrate to the graphene, not the pressure itself. The deformation of the substrate is induced by the pressure application. Therefore, the substrate deformation is $\varepsilon_S = \beta_S.\ P$, where β_S is the substrate linear compressibility and P is the applied pressure. Substrate deformation is transmitted, totally or partially depending on the substrate, yielding $\varepsilon_G = \alpha.\ \varepsilon_S$ with ε_G and ε_S the deformations of graphene and substrate, respectively. The parameter α corresponds then to a quantification of the adhesion degree between the 2D–system and its substrate. In the following, we refer to α as the adhesion parameter, with $\alpha = 1$ corresponding to perfect adhesion and $\alpha = 0$ to perfect sliding.

This relation leads to $\beta_G.\ \sigma = \alpha.\ \beta_S.\ P$, where β_G is the graphene linear bulk modulus and σ is the stress transmitted by the deformation of the substrate. This point is important as it clearly appears that *the stress on graphene is not equal to the applied pressure*. Once this point has been clarified, it implies that plotting the Raman shift as a function of pressure is not adapted. It is necessary to calculate the induced stress by $\sigma = \alpha \frac{\beta_S}{\beta_G}.P$. The value of β_G is taken as the in-plane linear bulk modulus of graphite, thus assuming that the reduction of dimensionality does not induce modifications of the C–C chemical bond and the associated mechanical properties. Through x-ray diffraction experiments on graphite, it has been determined that β_G = 1250 GPa.[23]

Figure 3a reports the collection of data with the G-peak position as a function of pressure for graphene sitting on several substrates, and Fig. 3b corresponds to the same data but as a function of the induced stress σ – using the linear compressibility of the different substrates shown in Table 1. After this rescaling, the data obtained for graphene on diamond, copper, and sapphire line up with the slope obtained in graphite, for which the Raman shift is physically due to the pressure effect on the chemical bonds.

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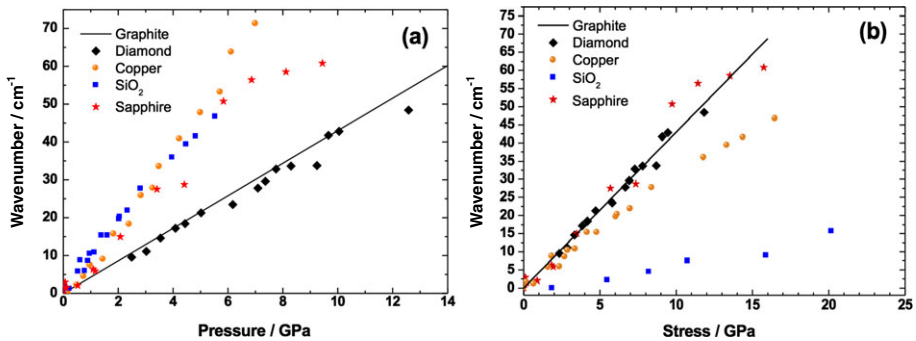

Figure 3. (a) Raman shift of the G-peak as a function of pressure for graphene single layer deposited on different substrates. The pressure-induced shifts in each case show some dispersion and are far from the value found in graphite despite the similar chemical bond. (b) Raman shift of the G-peak as a function of induced stress σ by the substrate deformation assuming a strain transfer function $\alpha = 1$ (see text). For all substrates but SiO_2, the same stress-dependency (~4.4 $cm^{-1}.GPa^{-1}$) is found in the low-stress range. In the case of copper, a change of regime is observed around σ = 3 GPa.

One notices that in the case of copper ($\sigma > 3 \pm 1$ GPa) and Si/SiO_2 substrates, the slope of $\omega_G(\sigma)$ is lower than that observed for graphite. This may be interpreted as a partial transmission of the deformation from the substrate to the graphene. At the interface, the graphene is under tension, whereas the substrate is under compression, thus leading to an interfacial elastic energy (mismatch). Whenever this energy is higher than the adhesion energy, a relaxation occurs through a slipping or wrinkling of graphene on the substrate (the relaxation process remains to be deeper investigated), and the strain is only partly transmitted, leading to $\alpha \neq 1$. This case is found when the pressure-dependence is different from the graphite one. The effectiveness of this transfer is indicated by the ratio of the measured slope of $\omega_G(\sigma)$ with the one measured for graphite (~4.4 $cm^{-1}.GPa^{-1}$) thus giving the $\alpha = 1$ reference.

Applying such method to SiO_2, one finds α ~0.2, and this value is constant over the whole pressure range. In the case of copper, above $\sigma = 3 \pm 1$ GPa, one can note a change of regime, from $\alpha = 1$ to α ~0.6, thus indicating a partial unbinding above this critical stress.

Using Fig. 4, in which α is represented as function of βs, it is possible to determine the substrate mechanical properties that are required to have a total strain transfer. It mostly corresponds to the change of regime when considering the slope of $\omega_G(\sigma)$ as function of the substrate linear bulk modulus. When the strain transfer is total, the slope corresponds to that of graphite. If the transfer is only partial, then the slope is lower. The critical value of substrate linear bulk modulus is thus βs ≈ 700 GPa. This result fixes the limit of validity of the total strain transfer between the substrate and graphene.

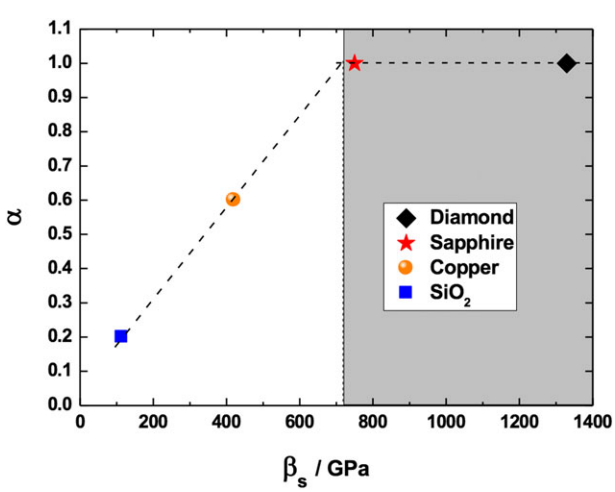

Figure 4. Strain transfer function between the substrate and graphene, α, as a function of the linear compressibility of the substrate, β_s. Total strain transfer is only possible for certain mechanical properties of the substrate. For $\beta_s < 700$ GPa, only a partial strain transfer may be expected.

We may note that in the case of a-SiO_2/Si, the 300 nm a-SiO_2 layer lies on a thick Si substrate having a bulk modulus roughly twice that of a-SiO_2. The amorphous nature of the a-SiO_2 layer should not allow for an epitaxial transmission of the constraint. We can then assume that it is the a-SiO_2 compressibility that is transmitted to the graphene layer. For much lower interface layer thicknesses (of the order of 10 nm) and in the case of an epitaxial layer, the silicon substrate compression could dominate.[36]

High-pressure response of graphene: effect of the number of layers

The pressure-dependence of the G-mode of few-layers graphene deposited on Si/SiO_2 substrate changes with the number of layers. In the case of monolayer and bilayer graphene, the slope $\frac{\partial \omega_G}{\partial P}$ is larger than the one obtained for trilayers and multilayers. Typically, the slope is around 8 $cm^{-1}.GPa^{-1}$ for graphene and bilayer, whereas it is close to the value found in graphite (~4.4 $cm^{-1}.GPa^{-1}$) for trilayers and more layers (Fig. 5).[30] Such observations may be explained considering the graphene systems as a membrane in adhesion with the substrate in which its bending rigidity is modified by increasing the number of graphene layers. As it was previously explained, it is not the pressure that is directly transmitted to the 2D material but a biaxial strain instead. As SiO_2 is highly compressible, leading to huge induced stresses σ, the dragging effect is only partial ($\alpha \neq 1$), and slipping phenomena can occur. Such effect is also observed for a bilayer but not for a trilayer graphene. A calculation taking into account the competition between the adhesion energy (gain) and bending energy (cost) shows that the unbinding event between a graphene stacking and a SiO_2 substrate occurs for a number of layers higher than two.[30] Therefore, the strain cannot be transmitted anymore through the substrate deformation and, for a trilayer graphene, the pressure is then directly transmitted to the sample from the pressure transmitting medium, and the pressure dependence of the G-band is similar to graphite.

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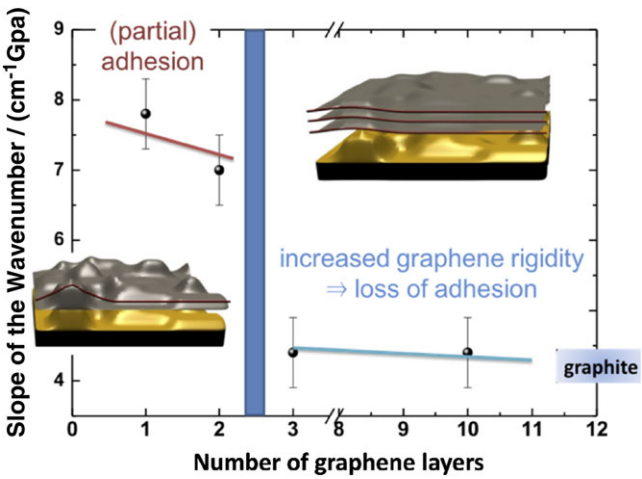

Figure 5. Pressure-dependence of the G-peak as a function of the number of graphene layer (n). For $n = 1$ (single layer) and $n = 2$ (bilayer), the slope is significantly higher than for samples with $n \geq 3$. The regime change for $n > 2$ is interpreted as a loss of adhesion related to the higher rigidity of the multilayer system as depicted by the images (see text).

High pressure response of graphene: effect of the pressure transmitting medium

Figure 6 shows our measured values of the G-band wavenumber pressure evolution of monolayer exfoliated graphene sitting on a SiO_2/Si substrate by using 4 : 1 methanol : ethanol and argon as PTM. The data is compared with our measurement of highly oriented pyrolytic graphite samples using argon as PTM. We should underline that only the alcohol PTM remains liquid in the pressure range shown here, while argon solidifies at about 1.4 GPa – but it is known to remain quasi-hydrostatic up to higher pressures.[24] In the particular case of argon, it is found that the first signs of pressure gradients appear from 2 GPa. As shown in Fig. 6, neither argon solidification nor its progressive loss of hydrostaticity appears to affect the G-band evolution of graphene with respect to the one in alcohol. This result appears to be in good agreement with the results of Proctor *et al.*[29] who did not observe qualitative differences for experiments performed on graphene using 4 : 1 methanol : ethanol or nitrogen as PTMs. We should note here that Nicolle *et al.*[30] have proposed a different result from what is shown in Fig. 1. In their case, the fitting of the G-band pressure slope was done in argon including the ambient pressure point and limited to pressures below the argon solidification point (very few points) that lead to a biased result and what we think an erroneous interpretation of a charge transfer effect between graphene and the substrate in the presence of alcohol PTM. As a matter of fact, Filintoglou *et al.*[31] did not observe any pressure-induced charge transfer for either polar or nonpolar PTMs, as their measurements for Cu-deposited graphene with 4 : 1 methanol : ethanol and Fluorinert PTMs gave G-peak slopes of 9.1 and 9.3 cm^{-1}/GPa – i.e. within error bars not showing any difference.

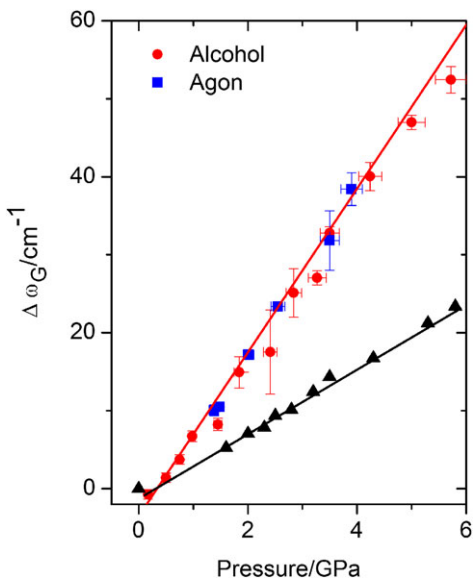

Figure 6. Pressure evolution of the G-band wavenumber of a single layer graphene sitting on a SiO_2/Si substrate in two different pressure transmitting media: a 4 : 1 methanol : ethanol (red circles) and argon (blue squares). The black triangles are our corresponding measurement for graphite using argon as pressure transmitting medium.

Figure 7 shows our data for exfoliated bilayer graphene supported on Si/SiO_2. Both the relative variation of the G-band position and of its full width at half maximum (FWHM) are represented with respect to pressure, for 4 : 1 methanol : ethanol, water or argon PTM. The data only show the compression ramp. A continuous line with a slope of 10 $cm^{-1}.GPa^{-1}$ corresponding to the data fit for monolayer graphene is included as reference. We observe that the peak evolution remains close to the one for monolayer graphene but shows anomalies that can be classified in different types. First of all, we observe a sudden softening of the G-band at 9 ± 1 and 7 ± 1 GPa in H_2O and argon PTMs, respectively. This softening is not observed in 4 : 1 methanol : ethanol up to the maximum pressure of 7 GPa. According to the general interpretation of the same behavior in monolayer graphene,[31] this softening could be related to an unbinding of the bilayer graphene system from its substrate. In the following, we will discuss the other features of the G-band pressure evolution. Except for 4 : 1 methanol : ethanol, which remains liquid in all the studied pressure range, we have included in Fig. 7 dashed vertical lines pointing to the solidification pressures of the PTM, its solid–solid transformation for the case of water, or its hydrostatic limit for the case of argon. Surprisingly, in the case of water and alcohol PTM, the pressure evolution of the G-band peak position and of its FWHM are quite similar to the ones for monolayer graphene. This is surprising because the alcohol mixture is a perfectly hydrostatic PTM in this pressure domain, whereas water is considered as a very bad PTM after its solidification. In both cases, the G-band slope is very close to the one for monolayer graphite as discussed in the previous section. The most significant effect is the particular evolution of the G-band FWHM above 2–3 GPa. This may be related to the well-known change of sign on the pressure derivative of the bulk modulus of a-SiO_2 at that pressure, but this hypothesis needs to be explored by studying bilayer graphene deposited on other substrates. Argon shows a more significant deviation in its evolution associated to its loss of hydrostatic behavior. To understand such difference of behavior, we include in Fig. 8 the pressure evolution of the surface of the PTM in contact with graphene compared with the one of the SiO_2 substrate. When considering its pressure derivative, it can be appreciated that ice follows a compression curve similar to the one of SiO_2, meaning that there is an absence of shear components in the case of ice compression. And this is not the case for solid-argon in which differences of compressibility between the substrate and the PTM persist. The stress asymmetry between the two graphene layers in argon would

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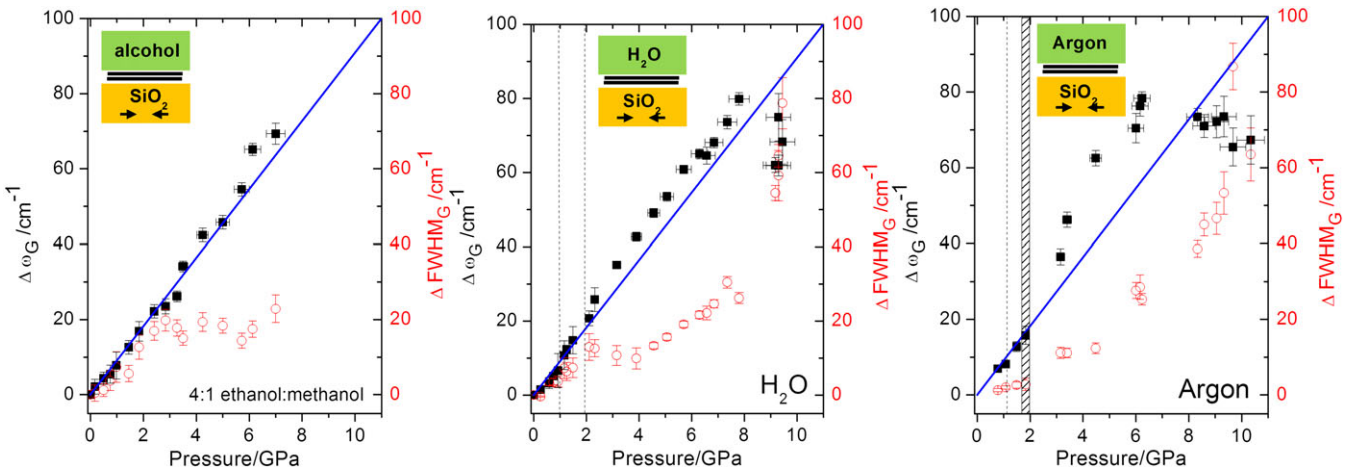

Figure 7. Pressure evolution of the G-band wavenumber variation (black points) and corresponding FWHM (red points, right-side scale) in exfoliated bilayer graphene on SiO_2/Si substrate immersed in three different transmission media: 4 : 1 methanol : ethanol, water and argon. The continuous lines in each panel are a reference for comparison corresponding to 9.1 $cm^{-1}GPa^{-1}$. The vertical lines correspond to different phase transitions of the pressure transmitting media and the hatched band in argon to its pressure recrystallization domain.

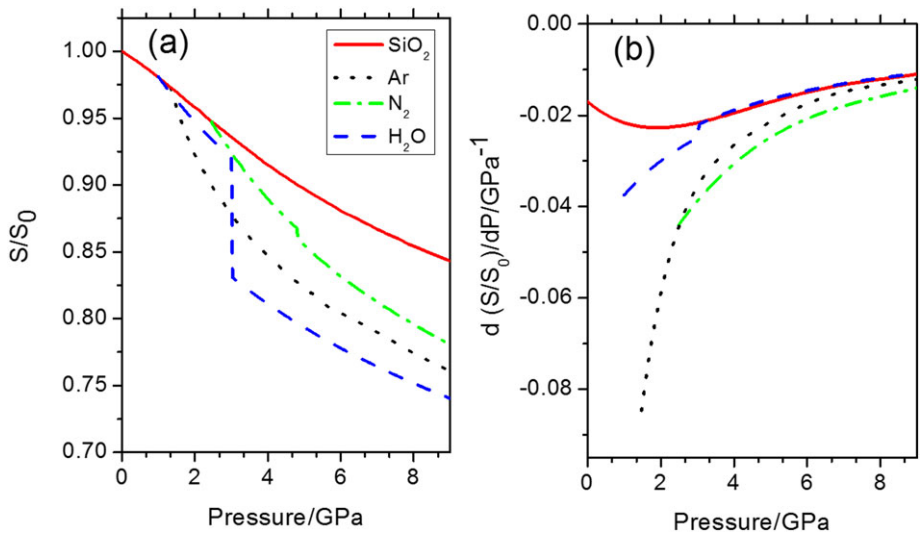

Figure 8. Pressure evolution of the surface of different systems (a) and its corresponding pressure derivative (b). Substrates and pressure transmitting media used in different studies are included.

then most likely be responsible of the observed deviations with respect to the graphene monolayer case. The different compressibility between the substrate and the PTM appears then as a fundamental aspect for the understanding of the mechanical response of graphene and of its corresponding pressure evolution of the Raman signal.

We should note that it has been recently proposed that bilayer graphene (supported on Teflon) under pressure in the presence of water as PTM leads to the formation of C–OH bonds and the consequent transformation to a new sp^3 structure named diamondene[6]. The observed slope of the G-band in that study would correspond in our interpretation to a fully unbound graphene system from its substrate. In any case, the eventual formation of C–OH bonds in our data (Fig. 7) does not show a strong modification of the G-band evolution with respect to the monolayer system except the already mentioned effect of crystallization. Consequently, carbon bonding will not have a significant effect on the evolution of graphene elastic constants. This could be modeled to corroborate the formation of the C–OH structures.

We consider now an important parameter affecting the sample evolution under pressure. It has been shown for bilayer graphene that in the absence of PTM, the Raman D-band – which characterizes the concentration of defects – strongly increases even for pressures below 5 GPa.[35] In Fig. 9, we show for monolayer graphene a comparison of the Raman spectra before and after a pressure cycle. Two different PTM were used, namely, argon and 4 : 1 methanol : ethanol with maximum reached pressure of 8 and 7 GPa, respectively. We can appreciate that no significant evolution of the D-band is observed for the alcohol PTM, whereas there is a strong increase of the D-band signal for argon PTM. Because the alcohol mixture remains liquid in that pressure range whereas argon solidifies at 2 GPa, we may conclude that defects formation may be related to the shear stress leading to the sample tearing.

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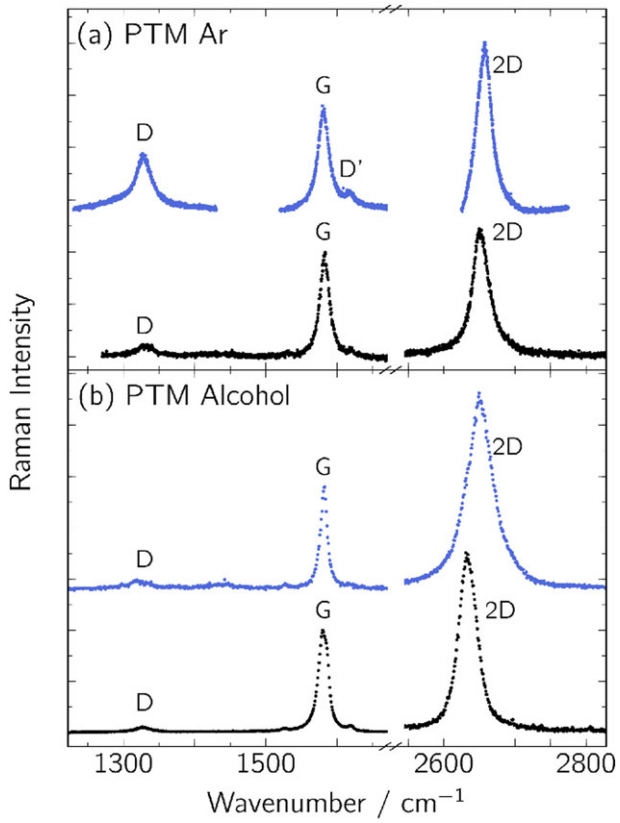

Figure 9. Effect of the pressure transmitting media (PTM) on the reversibility after a pressure cycle. Raman spectra of exfoliated pristine graphene monolayer supported on SiO_2/Si (black or bottom spectrum in each panel) and after a pressure cycle (blue or top spectrum in each panel) using two different PTM: argon (a) and 4 : 1 methanol : ethanol (b). The maximum reached pressure was 8 GPa (a) and 7 GPa (b).

Concluding remarks

In this manuscript, we have reviewed the high pressure Raman studies of graphene in the multi-GPa domain. We have situated high pressure studies in the forest of the mechanical studies of 2D materials underlining the important role played by the substrate in supported samples. We have discussed how high pressure experiments translate into compressive biaxial stress experiments with enhanced variable adhesion between graphene and its substrate. The substrate roughness, the bending rigidity of the sample, and the adhesive interaction between both systems are the fundamental parameters entering into play that govern how biaxial stress is transferred to the sample and determine the evolution of the wavenumber of the graphene Raman G-band signal. We have seen how these fundamental parameters can be tuned by either modifying the number of graphene layers, the nature of the substrate, or the nature of the pressure transmitting media. New original data corresponding to this last particular case allowed showing that the Raman response of supported graphene can be used as a highly sensitive probe to the pressure transmitting media change of phase, structure, or microstructure. Finally, the pressure-dependence of graphene may not differ strongly from that of graphite as far as no significant chemical interaction with the pressure transmitting medium takes place. We have shown that to understand the Raman response of graphene under pressure, there is a need of considering both the biaxial strain transmitted through the substrate compressibility and the adhesion/slipping evolution between substrate and graphene (given by the α parameter).

The concepts outlined in this manuscript can be obviously applied to other systems, as it has been recently done on the Raman study of supported MoS_2 under high pressure.[25]

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